MuonClip (Muon + QK-Clip)

Cap attention logits by rescaling Q/K weights after each step

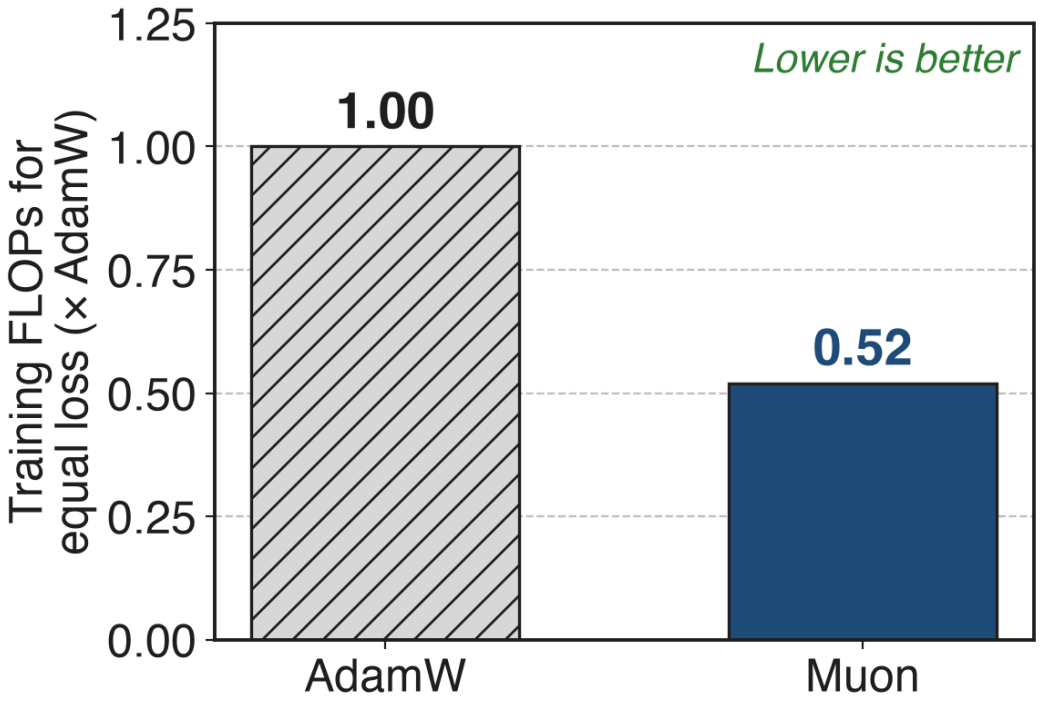
Moonshot AI · Kimi K2

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Optimizer & Stability · arxiv.org/abs/2507.20534

Muon matches AdamW with half the FLOPs

- Kimi K2: **1.04T** MoE, **32.6B** active
- **15.5T** tokens, all with Muon
- MLA attention: **61** layers × **64** heads



Vanilla Muon drives max logits past 1000

> 1000

max attention logit in a 53B-parameter MoE (9B active) trained with vanilla Muon, followed by loss spikes and occasional divergence

Kimi K2 report (arXiv 2507.20534), §2.1 and Fig. 2 (left)

Muon inflates the weights that set logits

(a) What bounds a logit

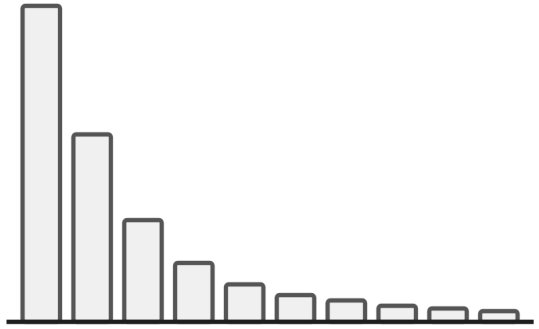
$|q_i \cdot k_j| \leq$ $\|x_i\| \, \|x_j\|$ $\|W_q\| \, \|W_k\|$

bounded by RMSNorm

spectral norms free to grow

$W_q W_k^T$ compounds growth

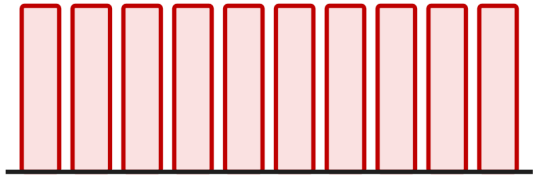
(b) Singular values of one update



Adam: few large

low effective rank

more weight directions can grow at once

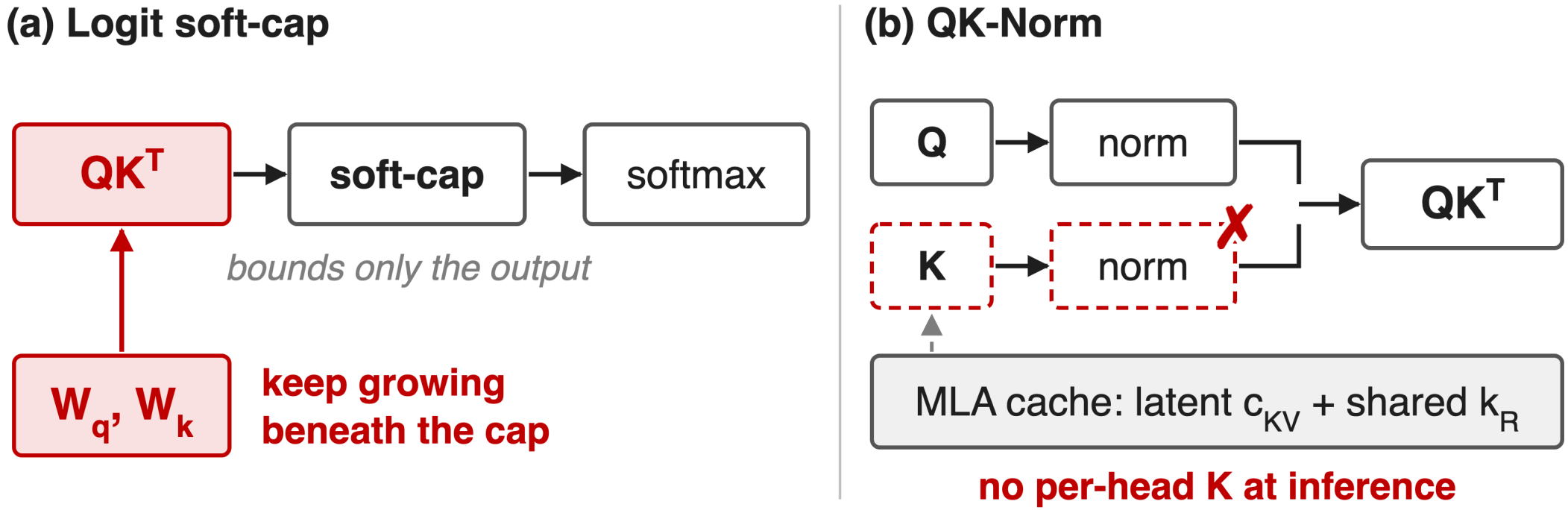


Muon: all equal

full rank

RMSNorm bounds the inputs, so the growth must come from `W_q` and `W_k`

Soft-cap and QK-Norm both fall short



Needed: a fix on the weights that works with MLA's latent keys

Insight: fix the weights, not the logits

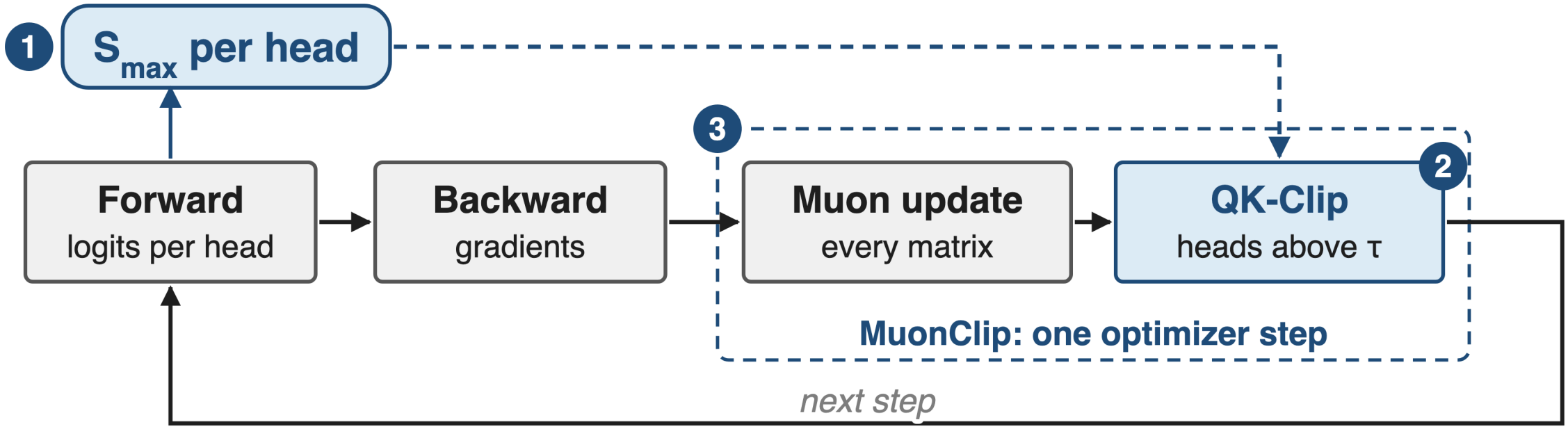
- Logit is bilinear: $\sqrt{\gamma}$ on `W_q` and `W_k` scales it by $\boldsymbol{\gamma}$
- Forward already computes each head's max logit

Insight: fix the weights, not the logits

- Logit is bilinear: $\sqrt{\gamma}$ on `W_q` and `W_k` scales it by $\boldsymbol{\gamma}$
- Forward already computes each head's max logit

After each update, rescale `W_q`, `W_k` only for heads above τ

MuonClip adds one clip after every Muon step

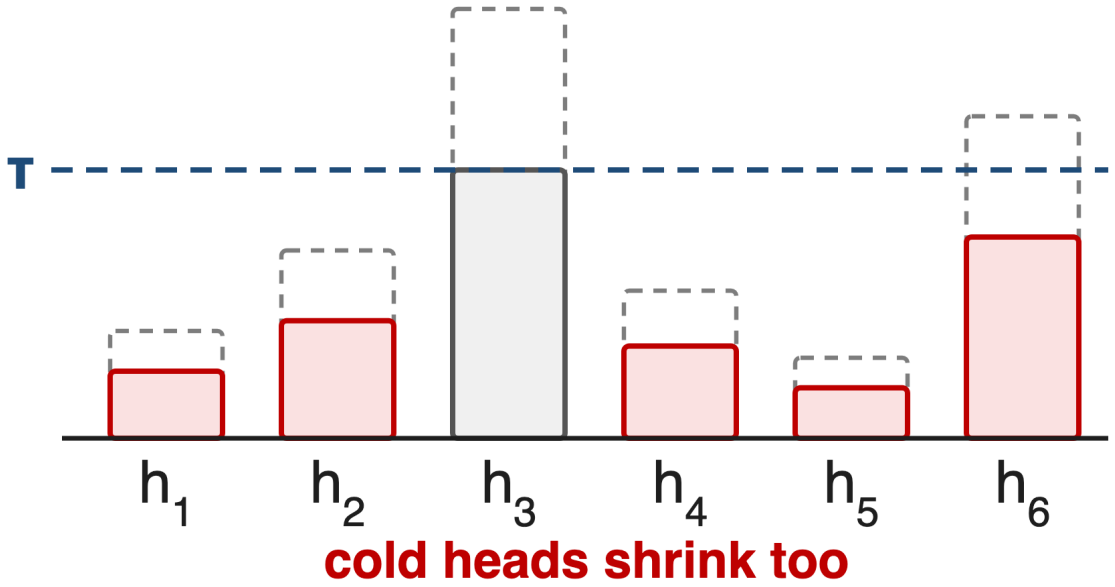


❶ per-head trigger · ❷ MLA-aware rescale · ❸ fused into the optimizer step

❶ Only heads above τ get clipped

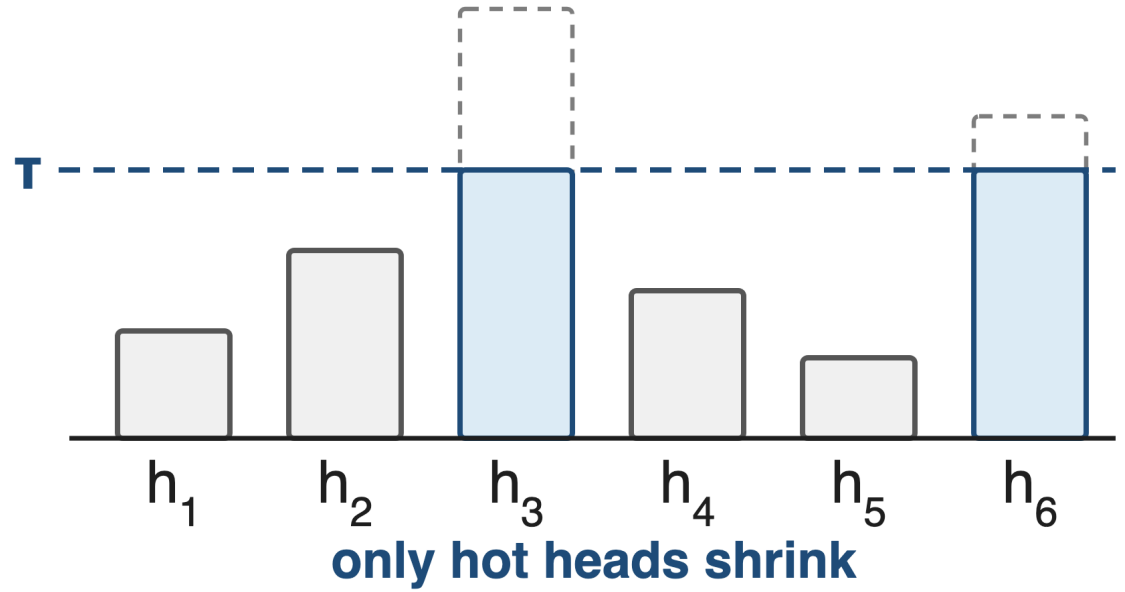
(a) Naive: one γ for all heads

$\gamma = \tau / \max_h S_{max}$



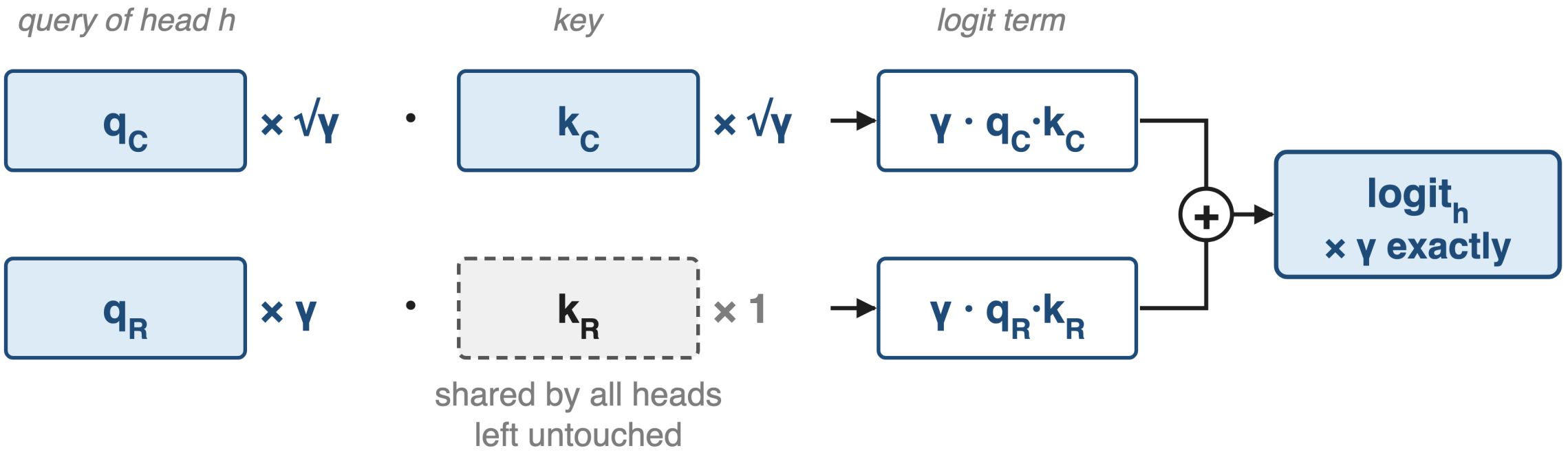
(b) QK-Clip: one γ per head

$\gamma_h = \min(1, \tau / S_{max}^h)$



Illustrative logits. Per-head clipping keeps cold heads untouched.

❷ MLA splits the scale: √γ, √γ, γ



Both terms shrink by γ; the shared k_R stays intact for other heads

❸ The clip runs inside the optimizer step

```
for each training step t:
  # Muon: every weight matrix W (n×m)
  M ← μ·M + G
  O ← NewtonSchulz(M) · 0.2·√max(n,m)
  W ← W − η·(O + λ·W)
  # QK-Clip: every attention head h
  if S_max[h] > τ:
    γ ← τ / S_max[h]
    W_qc ← √γ·W_qc;  W_kc ← √γ·W_kc
    W_qr ← γ·W_qr    # shared k_R untouched
```

update RMS matched
to AdamW: reuse its LR

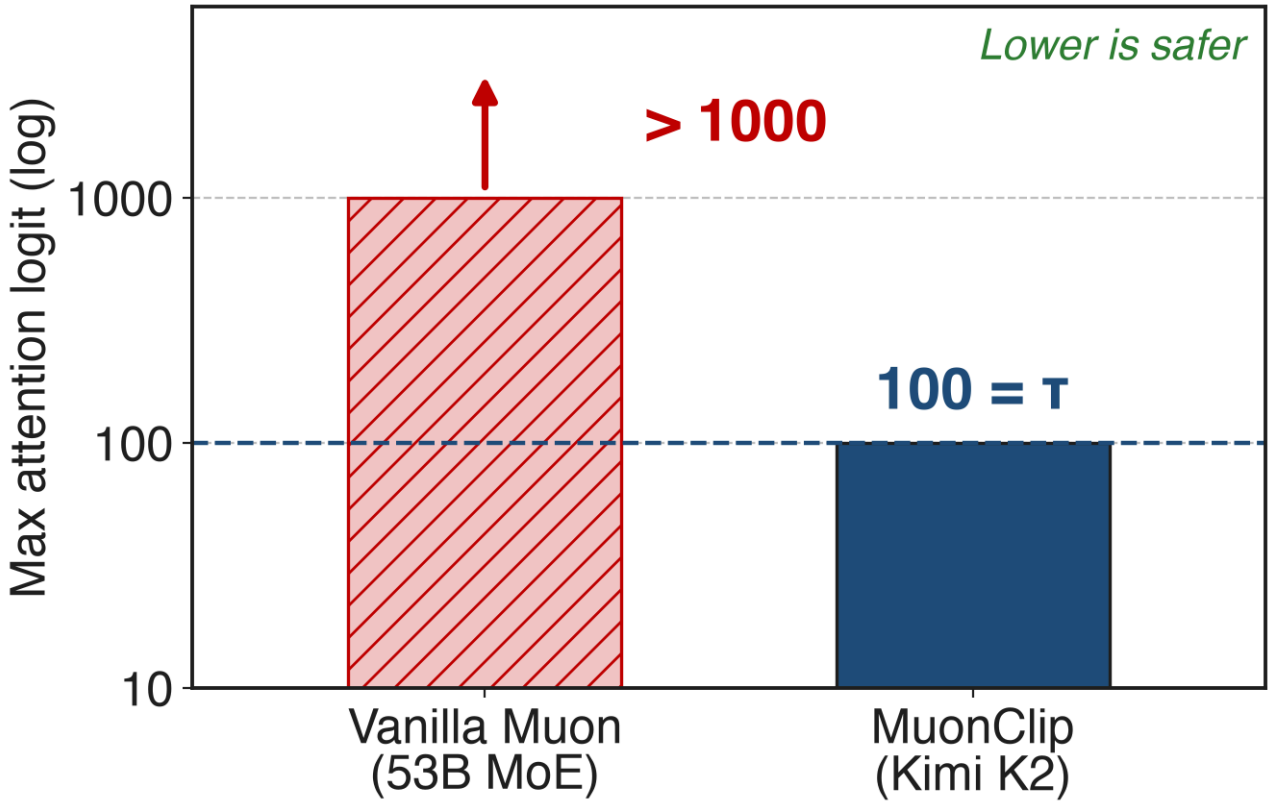
reuses the forward's
per-head max logit

runs after the update
this step's pass unchanged

One optimizer, MuonClip: Muon, weight decay, RMS matching, QK-Clip

Max logit capped at 100, not >1000

- K2 cap: **τ = 100**
- Logits decay below the cap after ~**30%** of steps
- No retuning of τ during the run



Different model scales; the vanilla bar is a lower bound

K2 trained 15.5T tokens with zero spikes

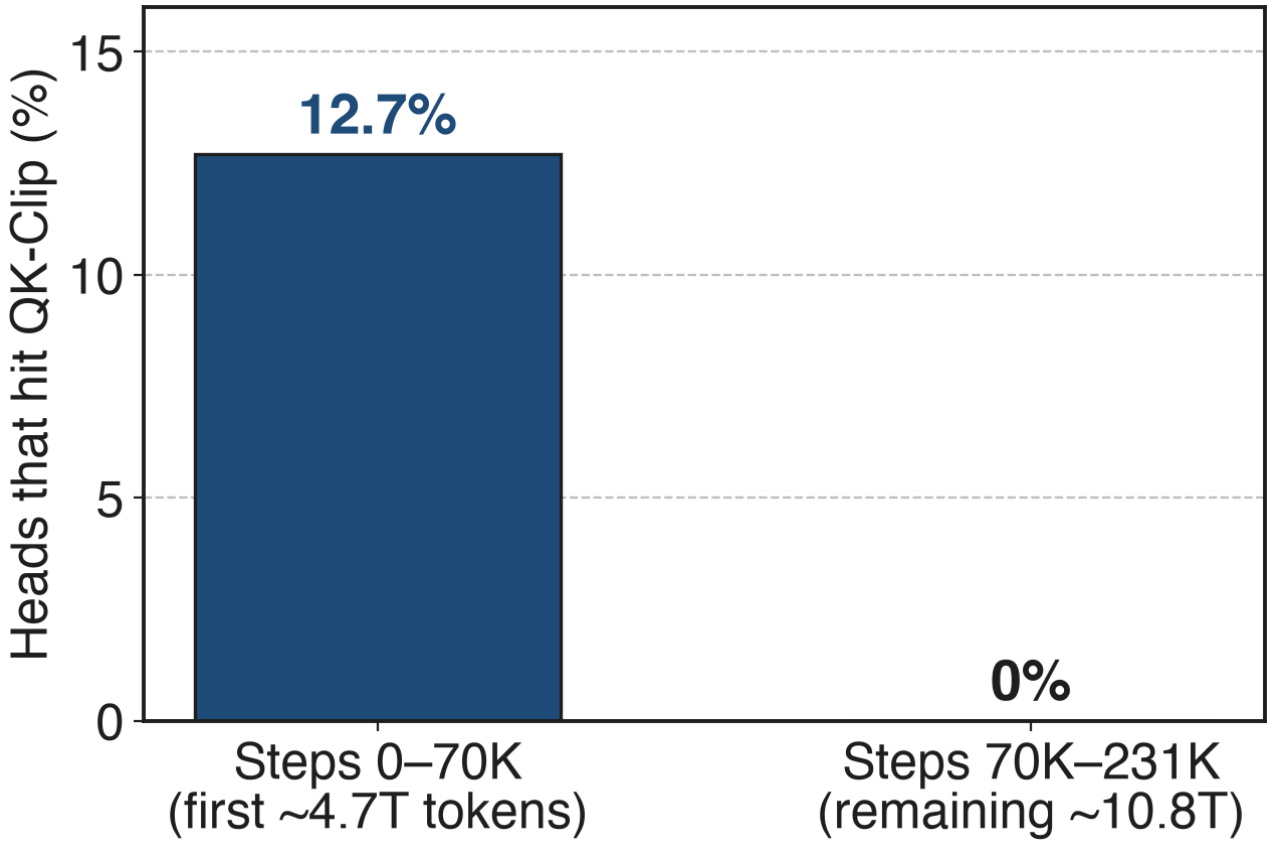
0

loss spikes in Kimi K2's 15.5T-token pretraining (per-step loss, no smoothing or sub-sampling)

Kimi K2 report (arXiv 2507.20534), abstract and Fig. 3

QK-Clip switched itself off after 70K steps

- Active early: **12.7%** of heads hit τ
- After 70K steps: all heads below τ, a no-op
- τ = **30** on a 3B MoE: negligible loss change



Step-to-token conversion uses the 67M-token batch (derived)

The cost: a threshold, a lag, MLA surgery

- **New knob τ**: 100 in K2, 30 in the ablation
- Acts from the next step only
- Needs each head's max over the global batch
- No-harm evidence: one 3B ablation, self-deactivation
- Non-MLA attention: QK-Norm is simpler (deck 66)
- K3 keeps the clip, adds per-head Muon

Summary

- Vanilla Muon: max attention logit **> 1000** at 53B scale
- **Insight**: logits are bilinear in `W_q`, `W_k`: rescale weights per head
- K2: capped at **τ = 100**, **12.7%** of heads clipped, then off
- **0** loss spikes across **15.5T** tokens on a **1.04T** MoE

https://arxiv.org/abs/2507.20534